

Pattern Matching and Machine Learning for Audio Signal Processing

Exercise sheet 9

To be uploaded in eCampus till: 12-07-2026 22:00 (strict deadline)

Exercise 9.1

[5 points]

Let $p \in [0, 1]$ denote the probability of a spectral peak to survive in a query audio fragment. We assume that this probability is independent and identically distributed over all points involved. In addition, we assume that the original document contains in each target zone exactly F spectral peaks. For each $k \in [1 : F]$ and a fixed anchor point compute the probability that this anchor point and at least k target points in its target zone survive. Simplify the formulas for $k = 1$ and $k = 2$. Compute for $p = 0.5$ and $F = 11$ this probability for $k = 2$.

(*Hint:* Interpret peak survival for a frame as a set of Bernoulli experiments.)

Exercise 9.2

[6 points]

Given an audio signal at a sampling rate of $F_s = 22050$ Hz, we want to compute a log-frequency spectrogram. As a requirement, all sets $P(p)$ for all pitches corresponding to the notes C2 ($p = 36$) to C3 ($p = 48$) should contain at least four Fourier coefficients. To meet this requirement, what is the minimal window length N (assuming that N is a power of two) to be used in the STFT? For this N , determine the elements of the set $P(36)$ explicitly.

Exercise 9.3

[4 + 4 = 8 points]

Let $\mathcal{F} = \mathbb{R}$ be the feature space and $c : \mathcal{F} \times \mathcal{F} \rightarrow \mathbb{R}_{\geq 0}$ the local cost measure defined by $c(x, y) = |x - y|$ for $x, y \in \mathbb{R}$. Compute DTW(X, Y) for the following sequences X and Y as well as **all** optimal warping paths. Also specify the cost matrix $\mathbf{C}$ and the accumulated cost matrix $\mathbf{D}$.

(a) $X = (1, 7, 4, 4, 6)$ and $Y = (1, 2, 2, 7)$.

(b) $X = (1, 2, 2, 1)$ and $Y = (1, 0, 0, 1)$.